Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: O'Banion Ranches

Physical address of dairy:

Number and Street	City	County	Zip Code
15775 S Indiana RD	Dos Palos	Merced	93620

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1936

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0085-0060-0009-0000 0085-0160-0009-0000

B. OPERATORS

O'Banion, Greg and Lisa

Operator name: O'Banion, Greg and Lisa

Telephone no.: (209) 587-1446 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
15735 Indiana RD	Dos Palos	CA	93620

This operator is responsible for paying permit fees.

C. OWNERS

O'Banion, Greg and Lisa

Legal owner name: O'Banion, Greg and Lisa

Telephone no.: (209) 587-1446 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
15735 Indiana RD	Dos Palos	CA	93620

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	1	0	100	225	0	0
Number under roof	0	0	0	0	0	0
Maximum number	1	0	100	230	0	0
Average number	0	0	100	225	0	0
Avg live weight (lbs)	1	0	1,200	770		

Predominant milk cow breed: Other

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 3,293.11 *tons per reporting period*

Total nitrogen from manure: 30,842.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 21,589.75 *lbs per reporting period*

Total phosphorus from manure: 5,219.50 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Surface water
Dairy	Ground water
Diesel Well	Ground water
Electric Well	Ground water
Feedlot 1	Ground water

Source Description	Type
Feedlot 2	Ground water
Shop Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/15/2023	Solid commercial fertilizer *60-20-29-9*	23.25 *ton*	As-is	0.1	6.000000	20.000000	29.000000	0.000000
05/10/2023	Solid commercial fertilizer *UN 32*	2.87 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000
07/01/2023	Solid commercial fertilizer *UN 32*	32.95 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	25,714.80	9,300.00	13,485.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	25,714.80	9,300.00	13,485.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	Corral solids	525.00 *ton*	Dry-weight	42.6		16,100.00	7,830.00	35,700.00		0.00
09/18/2023	Corral solids	1,475.00 *ton*	Dry-weight	23.2		18,900.00	4,900.00	27,900.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	52,523.31	15,820.58	84,726.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	52,523.31	15,820.58	84,726.63	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	27	27	1	none	X085-X270-X020-XXXX X085-X370-X008-XXXX
1a	35	35	1	none	X085-X270-X020-XXXX X085-X370-X008-XXXX
2	17	17	1	none	X085-X160-X006-XXXX
3	8	8	1	none	X085-X060-X010-XXXX X085-X070-X001-XXXX X085-X160-X006-XXXX
3a	7	7	1	none	X085-X060-X010-XXXX
4	13	13	1	none	X085-X060-X003-XXXX
5	25	25	1	none	X085-X060-X003-XXXX
6	11	11	1	none	X085-X060-X003-XXXX X085-X070-X001-XXXX
7	65	65	1	none	X085-X060-X003-XXXX X085-X070-X001-XXXX X085-X070-X025-XXXX
7a	11	11	1	none	X085-X060-X003-XXXX
8	56	56	1	none	X085-X060-X008-XXXX X085-X070-X016-XXXX
9	13	13	1	none	X085-X070-X016-XXXX
O'Banion Feedlot 1	18	18	1	none	X085-X030-X004-XXXX
O'Banion Feedlot 2	3	3	1	none	X085-X030-X004-XXXX
Totals for areas that were used for application					
Totals for areas that were not used for application	309	309	14		
Land application area totals	309	309	14		

B. CROPS AND HARVESTS

1

Field name: 1

1

11/01/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 27 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	189.00 *ton*	As-is		10.5	33,400.00	2,100.00	17,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.00	467.60	29.40	238.00	0.00

1a

Field name: 1a

10/01/2020: Pasture

Crop: Pasture Acres planted: 35 Plant date: 10/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	640.50 *ton*	As-is		71.4	5,200.00	800.00	6,900.00		100.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	256.50	0.00
Total actual harvest content	18.30	190.32	29.28	252.54	10,467.60

2

Field name: 2

2

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	459.00 *ton*	As-is		70.0	3,420.00	590.00	2,760.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.00	184.68	31.86	149.04	1,053.00

3

Field name: 3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	216.00 *ton*	As-is		70.0	3,420.00	590.00	2,760.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.00	184.68	31.86	149.04	1,053.00

3a

Field name: 3a

3a

09/20/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 7 Plant date: 09/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	49.00 *ton*	As-is		9.6	26,800.00	2,090.00	16,400.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.00	375.20	29.26	229.60	949.20

4

Field name: 4

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	312.00 *ton*	As-is		64.0	4,540.00	720.00	5,280.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.00	217.92	34.56	253.44	864.00

5

Field name: 5

5

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	600.00 *ton*	As-is		64.0	4,540.00	720.00	5,280.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.00	217.92	34.56	253.44	864.00

6

Field name: 6

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	264.00 *ton*	As-is		64.0	4,540.00	720.00	5,280.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.00	217.92	34.56	253.44	864.00

7

Field name: 7

7

09/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 65 Plant date: 09/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	520.00 *ton*	As-is		9.6	26,800.00	2,090.00	16,400.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.00	428.80	33.44	262.40	1,084.80

7a

Field name: 7a

10/01/2020: Pasture

Crop: Pasture Acres planted: 11 Plant date: 10/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	201.30 *ton*	As-is		71.4	5,200.00	800.00	6,900.00		100.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	256.50	0.00
Total actual harvest content	18.30	190.32	29.28	252.54	10,467.60

8

Field name: 8

8

09/20/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 56 Plant date: 09/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	448.00 *ton*	As-is		9.6	26,800.00	2,090.00	16,400.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.00	428.80	33.44	262.40	1,084.80

9

Field name: 9

06/08/2023: Sorghum

Crop: Sorghum Acres planted: 13 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	240.50 *ton*	As-is		70.0	5,370.00	871.00	6,550.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	168.00	48.00	144.00	0.00
Total actual harvest content	18.50	198.69	32.23	242.35	1,387.50

O'Banion Feedlot 1

Field name: O'Banion Feedlot 1

O'Banion Feedlot 1

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	504.00 *ton*	As-is		70.6	3,120.00	540.00	3,550.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.00	174.72	30.24	198.80	1,119.55

O'Banion Feedlot 2

Field name: O'Banion Feedlot 2

10/01/2022: Pasture

Crop: Pasture Acres planted: 3 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	54.90 *ton*	As-is		71.4	5,200.00	800.00	6,900.00		100.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	256.50	0.00
Total actual harvest content	18.30	190.32	29.28	252.54	10,467.60

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/01/2022: Alfalfa, hay

Field name: 1

Crop: Alfalfa, hay

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
60-20-29-9	Liquid commercial fertilizer	18.00	60.00	87.00	0.00	
Application event totals		18.00	60.00	87.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	Light rain	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	213.28	5,308,113.00 *gal*
Application event totals		0.00	0.00	0.00	213.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	182.25	4,535,846.00 *gal*
Application event totals		0.00	0.00	0.00	182.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	183.30	4,561,914.00 *gal*
Application event totals		0.00	0.00	0.00	183.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	218.91	5,448,229.00 *gal*
Application event totals		0.00	0.00	0.00	218.91	

1 - 11/01/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	184.34	4,587,982.00 *gal*
Application event totals		0.00	0.00	0.00	184.34	

1a - 10/01/2020: Pasture

Field name: 1a

Crop: Pasture

Plant date: 10/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	Light rain	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	215.43	6,950,402.00 *gal*
Application event totals		0.00	0.00	0.00	215.43	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	213.61	6,891,749.00 *gal*
Application event totals		0.00	0.00	0.00	213.61	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	170.89	5,513,399.00 *gal*
Application event totals		0.00	0.00	0.00	170.89	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	164.53	5,308,113.00 *gal*
Application event totals		0.00	0.00	0.00	164.53	

1a - 10/01/2020: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	168.16	5,425,419.00 *gal*
Application event totals		0.00	0.00	0.00	168.16	

2 - 06/01/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	216.84	3,397,974.00 *gal*
Application event totals		0.00	0.00	0.00	216.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	229.22	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	193.38	3,030,414.00 *gal*
Application event totals		229.22	0.00	0.00	193.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	242.92	3,806,591.00 *gal*
Application event totals		0.00	0.00	0.00	242.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	199.62	3,128,170.00 *gal*
Application event totals		0.00	0.00	0.00	199.62	

2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	199.62	3,128,170.00 *gal*
Application event totals		0.00	0.00	0.00	199.62	

3 - 06/01/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	Light rain	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	174.98	1,290,370.00 *gal*
Application event totals		0.00	0.00	0.00	174.98	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	229.22	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	139.63	1,029,689.00 *gal*
Application event totals		229.22	0.00	0.00	139.63	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	151.65	1,118,321.00 *gal*
Application event totals		0.00	0.00	0.00	151.65	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	185.59	1,368,574.00 *gal*
Application event totals		0.00	0.00	0.00	185.59	

3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	154.66	1,140,479.00 *gal*
Application event totals		0.00	0.00	0.00	154.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	201.49	1,485,881.00 *gal*
Application event totals		0.00	0.00	0.00	201.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	143.17	1,055,757.00 *gal*
Application event totals		0.00	0.00	0.00	143.17	

3a - 09/20/2022: Alfalfa, hay

Field name: 3a

Crop: Alfalfa, hay

Plant date: 09/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
60-20-29-9	Liquid commercial fertilizer	18.00	60.00	87.00	0.00	
Application event totals		18.00	60.00	87.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	Light rain	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	199.47	1,287,111.00 *gal*
Application event totals		0.00	0.00	0.00	199.47	

3a - 09/20/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	206.04	1,329,472.00 *gal*
Application event totals		0.00	0.00	0.00	206.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	206.04	1,329,472.00 *gal*
Application event totals		0.00	0.00	0.00	206.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	206.04	1,329,472.00 *gal*
Application event totals		0.00	0.00	0.00	206.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	138.87	896,090.00 *gal*
Application event totals		0.00	0.00	0.00	138.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	116.15	749,457.00 *gal*
Application event totals		0.00	0.00	0.00	116.15	

4 - 06/07/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	216.45	2,593,774.00 *gal*
Application event totals		0.00	0.00	0.00	216.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	179.47	2,150,617.00 *gal*
Application event totals		0.00	0.00	0.00	179.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	229.22	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	108.77	1,303,404.00 *gal*
Application event totals		229.22	0.00	0.00	108.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	97.89	1,173,064.00 *gal*
Application event totals		0.00	0.00	0.00	97.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	108.77	1,303,404.00 *gal*
Application event totals		0.00	0.00	0.00	108.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	103.33	1,238,234.00 *gal*
Application event totals		0.00	0.00	0.00	103.33	

4 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	195.78	2,346,127.00 *gal*
Application event totals		0.00	0.00	0.00	195.78	

5 - 06/07/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	225.11	5,187,548.00 *gal*
Application event totals		0.00	0.00	0.00	225.11	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.65	4,301,233.00 *gal*
Application event totals		0.00	0.00	0.00	186.65	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	229.22	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	113.12	2,606,808.00 *gal*
Application event totals		229.22	0.00	0.00	113.12	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	101.81	2,346,127.00 *gal*
Application event totals		0.00	0.00	0.00	101.81	

5 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	113.12	2,606,808.00 *gal*
Application event totals		0.00	0.00	0.00	113.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	107.46	2,476,468.00 *gal*
Application event totals		0.00	0.00	0.00	107.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	203.62	4,692,254.00 *gal*
Application event totals		0.00	0.00	0.00	203.62	

6 - 06/07/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	210.49	2,134,324.00 *gal*
Application event totals		0.00	0.00	0.00	210.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	212.10	2,150,617.00 *gal*
Application event totals		0.00	0.00	0.00	212.10	

6 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	229.22	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	213.39	2,163,651.00 *gal*
Application event totals		229.22	0.00	0.00	213.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	183.02	1,855,721.00 *gal*
Application event totals		0.00	0.00	0.00	183.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	150.24	1,523,353.00 *gal*
Application event totals		0.00	0.00	0.00	150.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	188.77	1,914,049.00 *gal*
Application event totals		0.00	0.00	0.00	188.77	

7 - 09/30/2021: Alfalfa, hay

Field name: 7

Crop: Alfalfa, hay

Plant date: 09/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
60-20-29-9	Liquid commercial fertilizer	18.00	60.00	87.00	0.00	
Application event totals		18.00	60.00	87.00	0.00	

7 - 09/30/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	234.94	14,076,763.00 *gal*
Application event totals		0.00	0.00	0.00	234.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	217.72	13,044,793.00 *gal*
Application event totals		0.00	0.00	0.00	217.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	182.28	10,921,222.00 *gal*
Application event totals		0.00	0.00	0.00	182.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	172.87	10,357,826.00 *gal*
Application event totals		0.00	0.00	0.00	172.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	190.23	11,397,942.00 *gal*
Application event totals		0.00	0.00	0.00	190.23	

7a - 10/01/2020: Pasture

Field name: 7a

Crop: Pasture

Plant date: 10/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7a - 10/01/2020: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	204.07	2,069,154.00 *gal*
Application event totals		0.00	0.00	0.00	204.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	165.44	1,677,481.00 *gal*
Application event totals		0.00	0.00	0.00	165.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	187.68	1,902,970.00 *gal*
Application event totals		0.00	0.00	0.00	187.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	165.82	1,681,391.00 *gal*
Application event totals		0.00	0.00	0.00	165.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	173.54	1,759,595.00 *gal*
Application event totals		0.00	0.00	0.00	173.54	

8 - 09/20/2022: Alfalfa, hay

Field name: 8

Crop: Alfalfa, hay

Plant date: 09/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

8 - 09/20/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
60-20-29-9	Liquid commercial fertilizer	18.00	60.00	87.00	0.00	
Application event totals		18.00	60.00	87.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	202.33	10,444,502.00 *gal*
Application event totals		0.00	0.00	0.00	202.33	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	189.37	9,775,530.00 *gal*
Application event totals		0.00	0.00	0.00	189.37	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	192.34	9,928,680.00 *gal*
Application event totals		0.00	0.00	0.00	192.34	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	185.59	9,580,019.00 *gal*
Application event totals		0.00	0.00	0.00	185.59	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	172.96	8,928,317.00 *gal*
Application event totals		0.00	0.00	0.00	172.96	

9 - 06/08/2023: Sorghum

9 - 06/08/2023: Sorghum

Field name: 9

Crop: Sorghum

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	141.10	0.00	0.00	0.00	
Application event totals		141.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	193.34	2,316,801.00 *gal*
Application event totals		0.00	0.00	0.00	193.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	165.87	1,987,691.00 *gal*
Application event totals		0.00	0.00	0.00	165.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	225.15	2,698,046.00 *gal*
Application event totals		0.00	0.00	0.00	225.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	228.82	2,742,036.00 *gal*
Application event totals		0.00	0.00	0.00	228.82	

O'Banion Feedlot 1 - 06/01/2023: Corn, silage

Field name: O'Banion Feedlot 1

Crop: Corn, silage

Plant date: 06/01/2023

O'Banion Feedlot 1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	170.86	2,834,904.00 *gal*
Application event totals		0.00	0.00	0.00	170.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	229.22	0.00	0.00	0.00	
CCID	Surface water	0.00	0.00	0.00	186.18	3,089,067.00 *gal*
Application event totals		229.22	0.00	0.00	186.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	182.64	3,030,414.00 *gal*
Application event totals		0.00	0.00	0.00	182.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	185.00	3,069,516.00 *gal*
Application event totals		0.00	0.00	0.00	185.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	153.18	2,541,638.00 *gal*
Application event totals		0.00	0.00	0.00	153.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.18	3,089,067.00 *gal*
Application event totals		0.00	0.00	0.00	186.18	

O'Banion Feedlot 1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	185.00	3,069,516.00 *gal*
Application event totals		0.00	0.00	0.00	185.00	

O'Banion Feedlot 2 - 10/01/2022: Pasture

Field name: O'Banion Feedlot 2

Crop: Pasture

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	200.32	553,947.00 *gal*
Application event totals		0.00	0.00	0.00	200.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	131.97	364,953.00 *gal*
Application event totals		0.00	0.00	0.00	131.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	120.19	332,368.00 *gal*
Application event totals		0.00	0.00	0.00	120.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	176.75	488,777.00 *gal*
Application event totals		0.00	0.00	0.00	176.75	

O'Banion Feedlot 2 - 10/01/2022: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	113.12	312,817.00 *gal*
Application event totals		0.00	0.00	0.00	113.12	

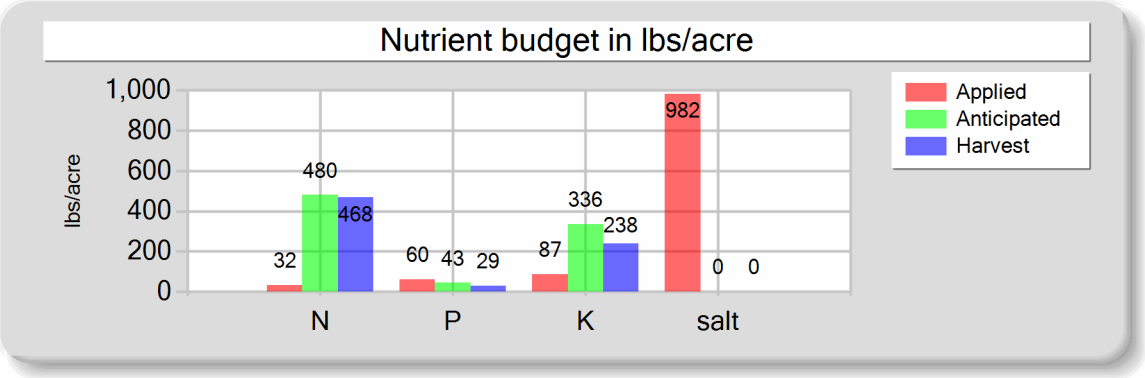
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	164.97	456,191.00 *gal*
Application event totals		0.00	0.00	0.00	164.97	

B. NUTRIENT BUDGET

1 - 11/01/2022: Alfalfa, hay

Field name: 1 Crop: Alfalfa, hay Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18.00	60.00	87.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	982.07
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	32.00	60.00	87.00	982.07
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	467.60	29.40	238.00	0.00
Nutrient balance	-435.60	30.60	-151.00	982.07
Applied to removed ratio	0.07	2.04	0.37	0.00

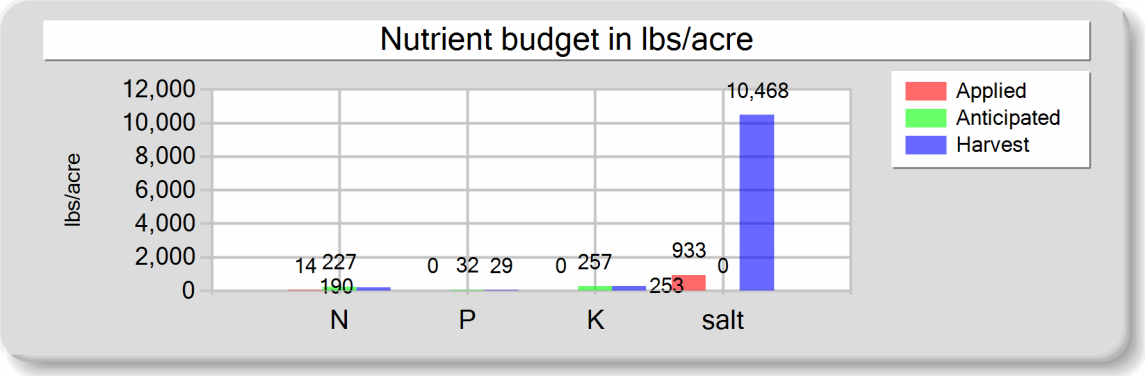
Fresh water applied	
24,442,084.00	*gallons*
900.12	*acre-inches*
33.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

1a - 10/01/2020: Pasture

Field name: 1a | Crop: Pasture | Plant date: 10/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	932.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	932.63
Anticipated crop nutrient removal	226.50	31.50	256.50	0.00
Actual crop nutrient removal	190.32	29.28	252.54	10,467.60
Nutrient balance	-176.32	-29.28	-252.54	-9,534.97
Applied to removed ratio	0.07	0.00	0.00	0.09

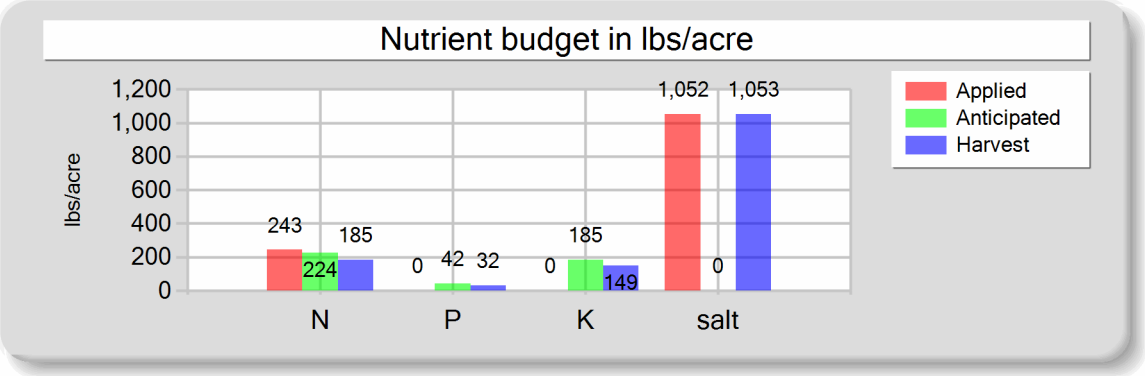
Fresh water applied	
30,089,082.00	*gallons*
1,108.08	*acre-inches*
31.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/01/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.22	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,052.39
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	243.22	0.00	0.00	1,052.39
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	184.68	31.86	149.04	1,053.00
Nutrient balance	58.54	-31.86	-149.04	-0.61
Applied to removed ratio	1.32	0.00	0.00	1.00

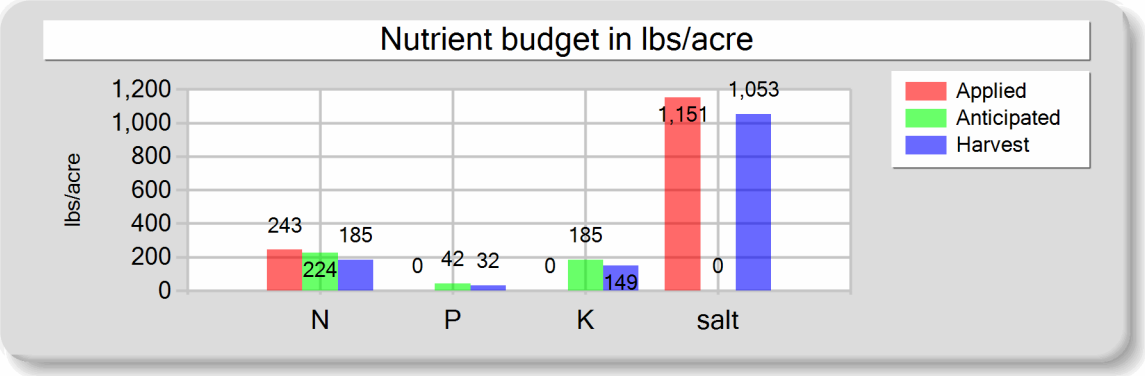
Fresh water applied	
16,491,319.00	*gallons*
607.32	*acre-inches*
35.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/01/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.22	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,151.17
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	243.22	0.00	0.00	1,151.17
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	184.68	31.86	149.04	1,053.00
Nutrient balance	58.54	-31.86	-149.04	98.17
Applied to removed ratio	1.32	0.00	0.00	1.09

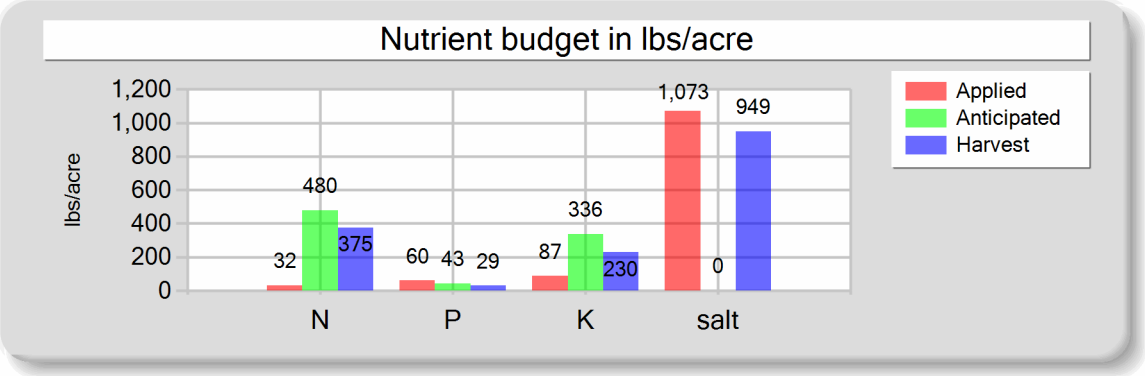
Fresh water applied	
8,489,071.00	*gallons*
312.62	*acre-inches*
39.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3a - 09/20/2022: Alfalfa, hay

Field name: 3a Crop: Alfalfa, hay Plant date: 09/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18.00	60.00	87.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,072.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	32.00	60.00	87.00	1,072.62
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	375.20	29.26	229.60	949.20
Nutrient balance	-343.20	30.74	-142.60	123.42
Applied to removed ratio	0.09	2.05	0.38	1.13

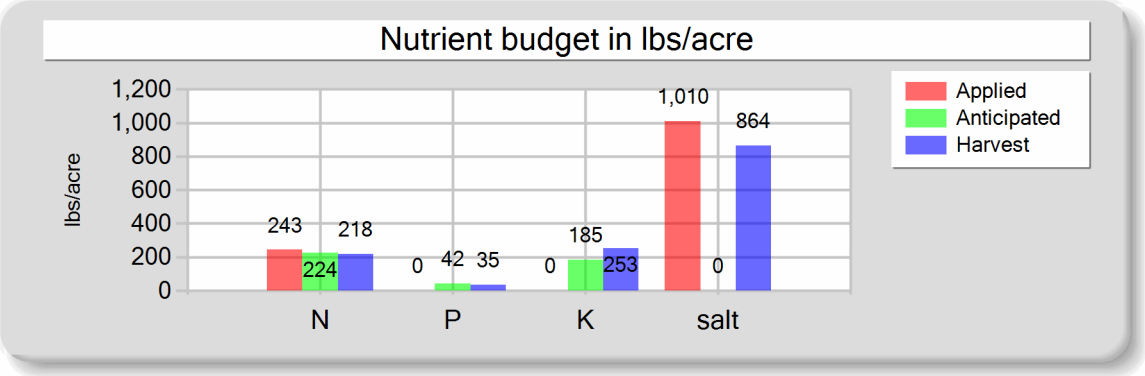
Fresh water applied	
6,921,074.00	*gallons*
254.88	*acre-inches*
36.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/07/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.22	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,010.46
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	243.22	0.00	0.00	1,010.46
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	217.92	34.56	253.44	864.00
Nutrient balance	25.30	-34.56	-253.44	146.46
Applied to removed ratio	1.12	0.00	0.00	1.17

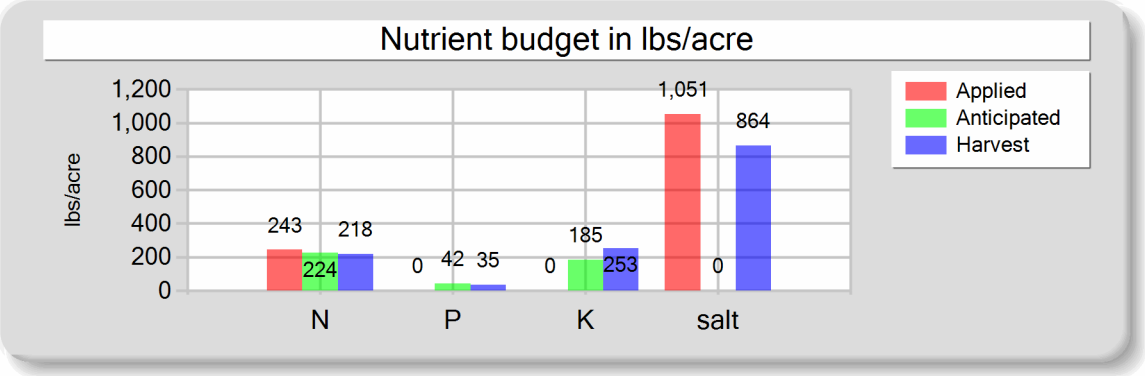
Fresh water applied	
12,108,624.00	*gallons*
445.92	*acre-inches*
34.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/07/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.22	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,050.88
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	243.22	0.00	0.00	1,050.88
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	217.92	34.56	253.44	864.00
Nutrient balance	25.30	-34.56	-253.44	186.88
Applied to removed ratio	1.12	0.00	0.00	1.22

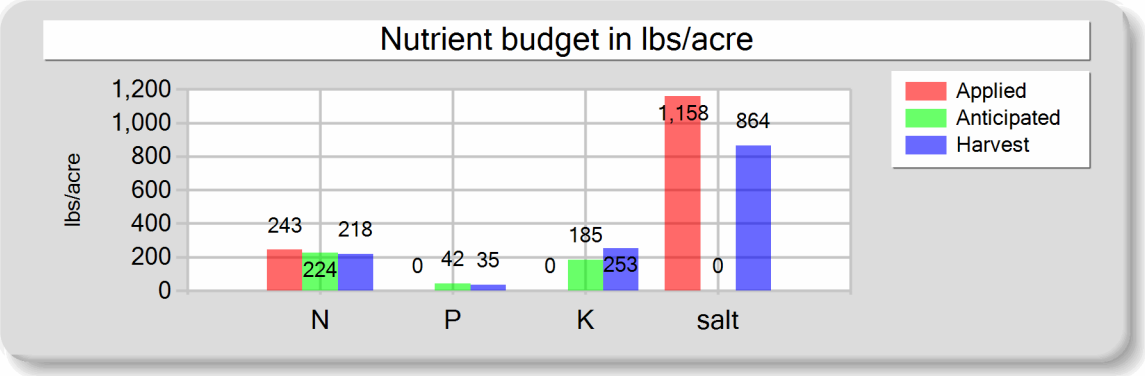
Fresh water applied	
24,217,246.00	*gallons*
891.84	*acre-inches*
35.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/07/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.22	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,158.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	243.22	0.00	0.00	1,158.00
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	217.92	34.56	253.44	864.00
Nutrient balance	25.30	-34.56	-253.44	294.00
Applied to removed ratio	1.12	0.00	0.00	1.34

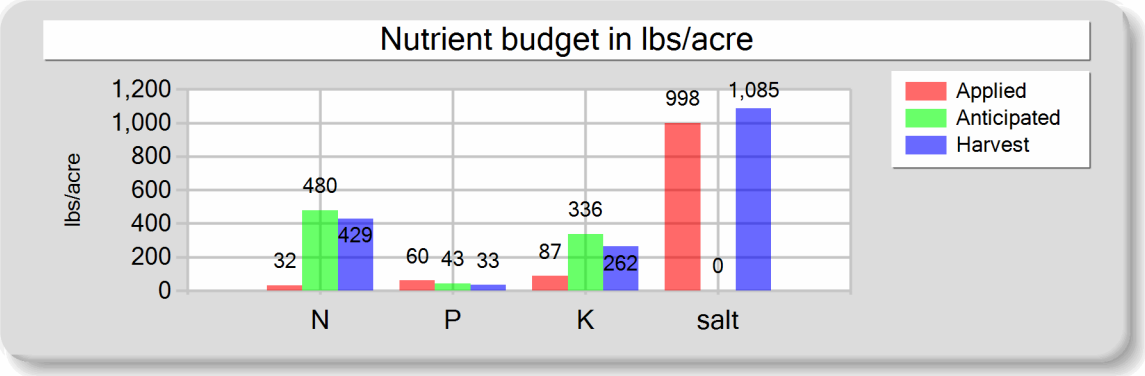
Fresh water applied	
11,741,715.00	*gallons*
432.41	*acre-inches*
39.31	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 09/30/2021: Alfalfa, hay

Field name: 7 Crop: Alfalfa, hay Plant date: 09/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18.00	60.00	87.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	998.04
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	32.00	60.00	87.00	998.04
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	428.80	33.44	262.40	1,084.80
Nutrient balance	-396.80	26.56	-175.40	-86.76
Applied to removed ratio	0.07	1.79	0.33	0.92

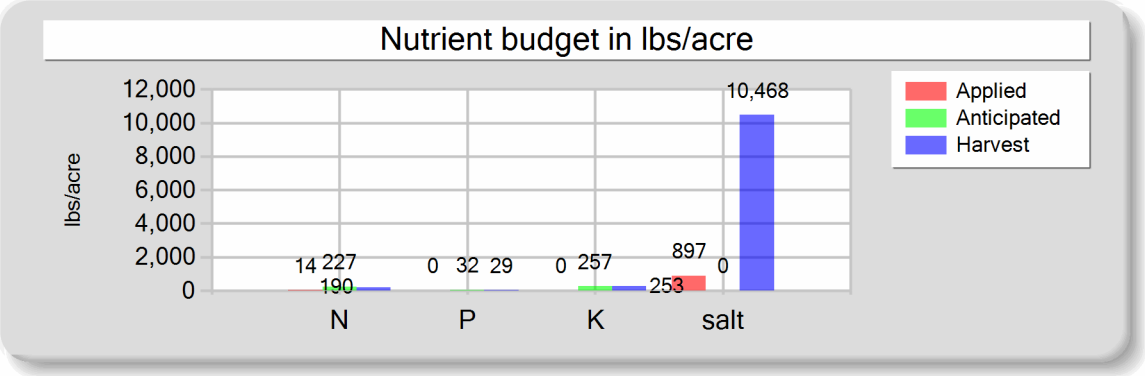
Fresh water applied	
59,798,546.00	*gallons*
2,202.18	*acre-inches*
33.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7a - 10/01/2020: Pasture

Field name: 7a | Crop: Pasture | Plant date: 10/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	896.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	896.54
Anticipated crop nutrient removal	226.50	31.50	256.50	0.00
Actual crop nutrient removal	190.32	29.28	252.54	10,467.60
Nutrient balance	-176.32	-29.28	-252.54	-9,571.06
Applied to removed ratio	0.07	0.00	0.00	0.09

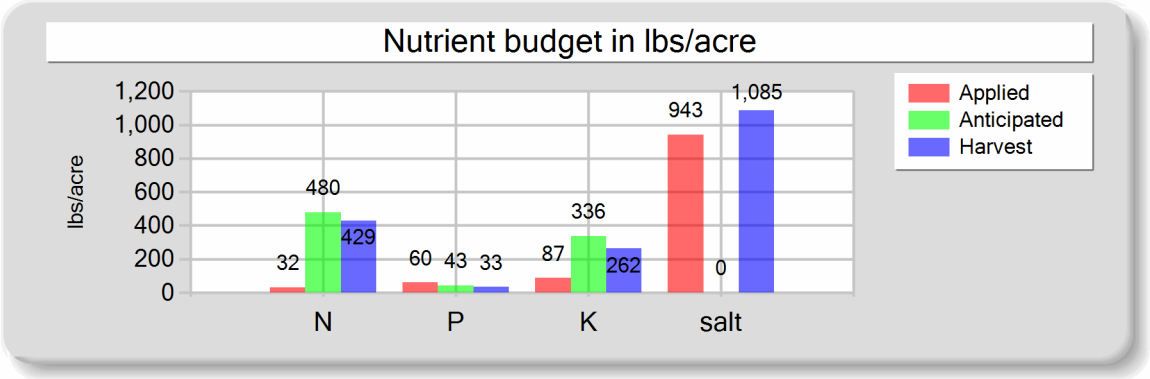
Fresh water applied	
9,090,591.00	*gallons*
334.78	*acre-inches*
30.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 09/20/2022: Alfalfa, hay

Field name: 8 Crop: Alfalfa, hay Plant date: 09/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18.00	60.00	87.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	942.60
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	32.00	60.00	87.00	942.60
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	428.80	33.44	262.40	1,084.80
Nutrient balance	-396.80	26.56	-175.40	-142.20
Applied to removed ratio	0.07	1.79	0.33	0.87

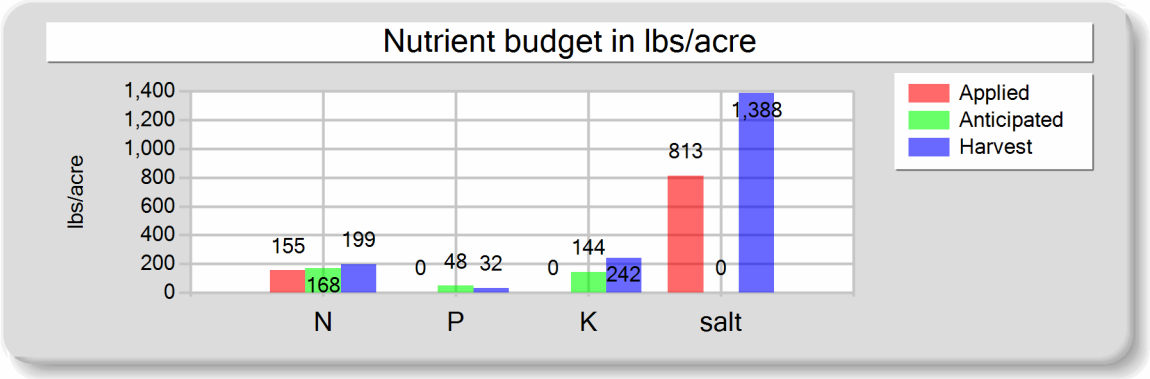
Fresh water applied	
48,657,048.00	*gallons*
1,791.87	*acre-inches*
32.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 06/08/2023: Sorghum

Field name: 9 Crop: Sorghum Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	141.10	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	813.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	155.10	0.00	0.00	813.18
Anticipated crop nutrient removal	168.00	48.00	144.00	0.00
Actual crop nutrient removal	198.69	32.23	242.35	1,387.50
Nutrient balance	-43.59	-32.23	-242.35	-574.32
Applied to removed ratio	0.78	0.00	0.00	0.59

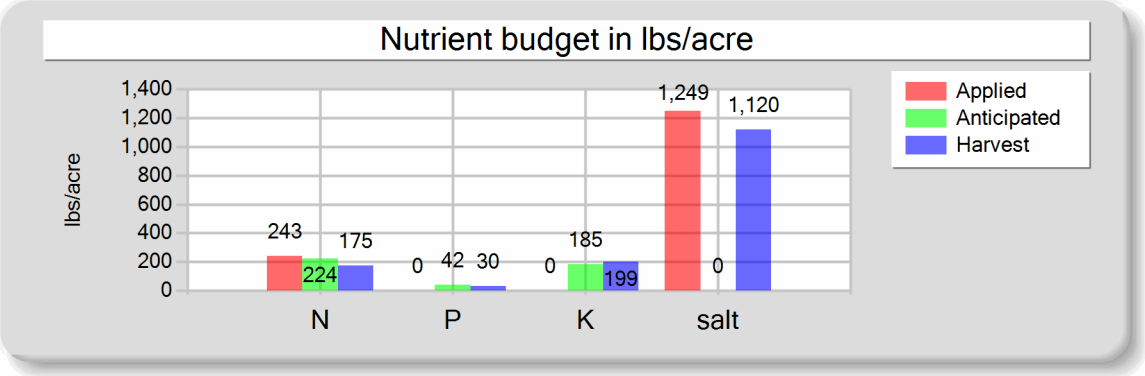
Fresh water applied	
9,744,574.00	*gallons*
358.86	*acre-inches*
27.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

O'Banion Feedlot 1 - 06/01/2023: Corn, silage

Field name: O'Banion Feedlot 1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.22	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,249.03
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	243.22	0.00	0.00	1,249.03
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	174.72	30.24	198.80	1,119.55
Nutrient balance	68.50	-30.24	-198.80	129.48
Applied to removed ratio	1.39	0.00	0.00	1.12

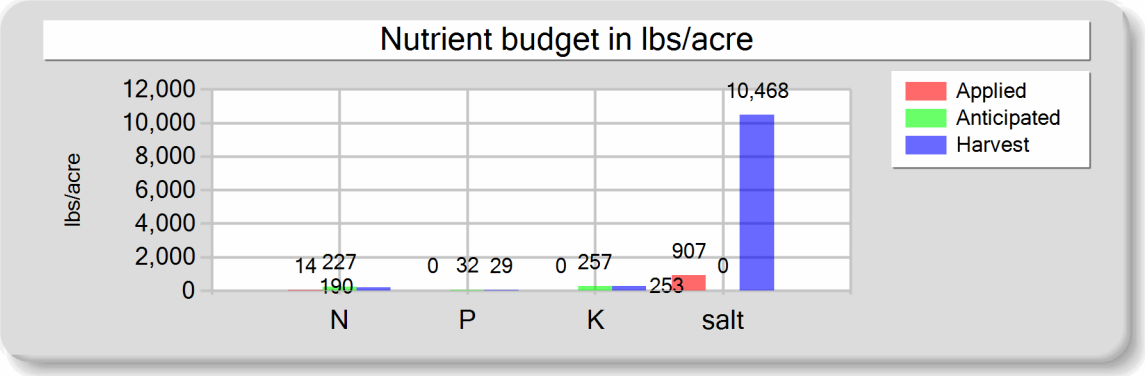
Fresh water applied	
20,724,122.00	*gallons*
763.20	*acre-inches*
42.40	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

O'Banion Feedlot 2 - 10/01/2022: Pasture

Field name: O'Banion Feedlot 2 Crop: Pasture Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	907.32
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	907.32
Anticipated crop nutrient removal	226.50	31.50	256.50	0.00
Actual crop nutrient removal	190.32	29.28	252.54	10,467.60
Nutrient balance	-176.32	-29.28	-252.54	-9,560.28
Applied to removed ratio	0.07	0.00	0.00	0.09

Fresh water applied	
2,509,053.00	*gallons*
92.40	*acre-inches*
30.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure @ Dairy

Sample and source description: Manure @ Dairy

Sample date: 08/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	7,830.00	35,700.00							
DL	100.00	100.00	100.00							

Manure @ Feedlot

Sample and source description: Manure @ Feedlot

Sample date: 08/25/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 23.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	4,900.00	27,900.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

CCID

CCID

Santa Rita@ Endworks

Sample description: Santa Rita@ Endworks

Sample date: 05/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00								81.00	130
DL		0.50	0.10								1.00	10

Dairy

Dairy

Sample description: Dairy

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00	53.00	24.40	118.00	180.00	0.00	112.00	161.00	1,050.00	525
DL		0.50	0.14	0.50	0.50	0.50	0.50	2.00	0.60	0.60	1.00	10

Electric Well

Well-Rocha

Sample description: Well-Rocha

Sample date: 06/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00		35.10	148.00	178.00	0.00	102.00	143.00	977.00	470
DL		0.50	0.14		0.50	0.50	0.50	2.00	0.60	0.50	100.00	10

Feedlot 1

Feedlot 1

Feedlot 1

Sample description: Feedlot 1

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.32	159.00	47.90	186.00	260.00	0.00	185.00	514.00	2,080.00	1,190
DL		0.50	0.14	0.50	0.50	0.50	0.50	2.00	0.60	0.50	1.00	10

Feedlot 2

Feedlot 2

Sample description: Feedlot 2

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00	10.40	1.07	45.30	94.00	0.00	0.00	20.50	266.00	120
DL		0.50	0.14	0.50	0.50	0.50	0.50	2.00	0.60	0.50	1.00	10

Shop Well

Shop Well

Sample description: Shop Well

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00	27.00	10.60	127.00	182.00	0.00	69.20	114.00	804.00	395
DL		0.50	0.14	0.50	0.50	0.50	0.50	2.00	0.60	0.50	1.00	10

D. SOIL ANALYSES

8

8

Sample and source description: 8

Sample date: 10/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3.00				
DL			0.50				

E. PLANT TISSUE ANALYSES

1 - 11/01/2022: Alfalfa, hay

Alfalfa Hay, G-O 6th 10/6/23

Sample and source description: Alfalfa Hay, G-O 6th 10/6/23

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,400.00	2,100.00	17,000.00		
DL	100.00	100.00	100.00		

1a - 10/01/2020: Pasture

Pasture

Sample and source description: Pasture

Sample date: 03/28/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	6,900.00		7.90
DL	100.00	100.00	100.00		0.01

2 - 06/01/2023: Corn, silage

Camp

Sample and source description: Camp

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	590.00	2,760.00		6.50
DL	100.00	100.00	100.00		0.01

3 - 06/01/2023: Corn, silage

Camp

Sample and source description: Camp

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	590.00	2,760.00		6.50
DL	100.00	100.00	100.00		0.01

3a - 09/20/2022: Alfalfa, hay

Jim 3 Rocha 7 Market 8

Sample and source description: Jim 3 Rocha 7 Market 8

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	2,090.00	16,400.00		7.50
DL	100.00	100.00	100.00		0.01

4 - 06/07/2023: Corn, silage

4,5,6

Sample and source description: 4,5,6

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,540.00	720.00	5,280.00		5.00
DL	100.00	100.00	100.00		0.01

5 - 06/07/2023: Corn, silage

4,5,6

Sample and source description: 4,5,6

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,540.00	720.00	5,280.00		5.00
DL	100.00	100.00	100.00		0.01

6 - 06/07/2023: Corn, silage

4,5,6

Sample and source description: 4,5,6

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,540.00	720.00	5,280.00		5.00
DL	100.00	100.00	100.00		0.01

7 - 09/30/2021: Alfalfa, hay

Jim 3 Rocha 7 Market 8

Sample and source description: Jim 3 Rocha 7 Market 8

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	2,090.00	16,400.00		7.50
DL	100.00	100.00	100.00		0.01

7a - 10/01/2020: Pasture

Pasture

Sample and source description: Pasture

Sample date: 03/28/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	6,900.00		7.90
DL	100.00	100.00	100.00		0.01

8 - 09/20/2022: Alfalfa, hay

Jim 3 Rocha 7 Market 8

Sample and source description: Jim 3 Rocha 7 Market 8

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	2,090.00	16,400.00		7.50
DL	100.00	100.00	100.00		0.01

9 - 06/08/2023: Sorghum

Sorghum

Sample and source description: Sorghum

Sample date: 09/20/2023 Source of analysis: Other/estimated Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,370.00	871.00	6,550.00		12.50
DL	100.00	100.00	100.00		0.01

O'Banion Feedlot 1 - 06/01/2023: Corn, silage

Feedlot

Sample and source description: Feedlot

Sample date: 09/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,120.00	540.00	3,550.00		6.80
DL	100.00	100.00	100.00		0.01

O'Banion Feedlot 2 - 10/01/2022: Pasture

Pasture

Sample and source description: Pasture

Sample date: 03/28/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	6,900.00		7.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

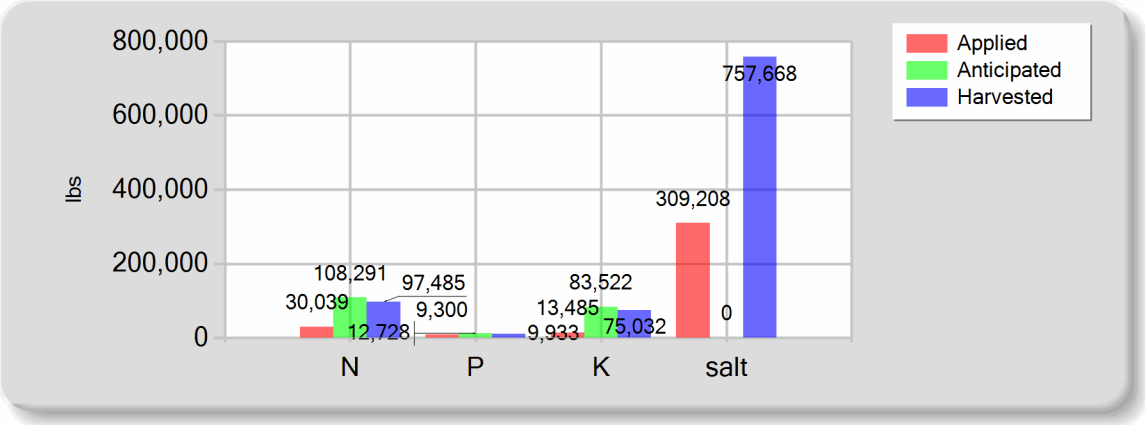
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

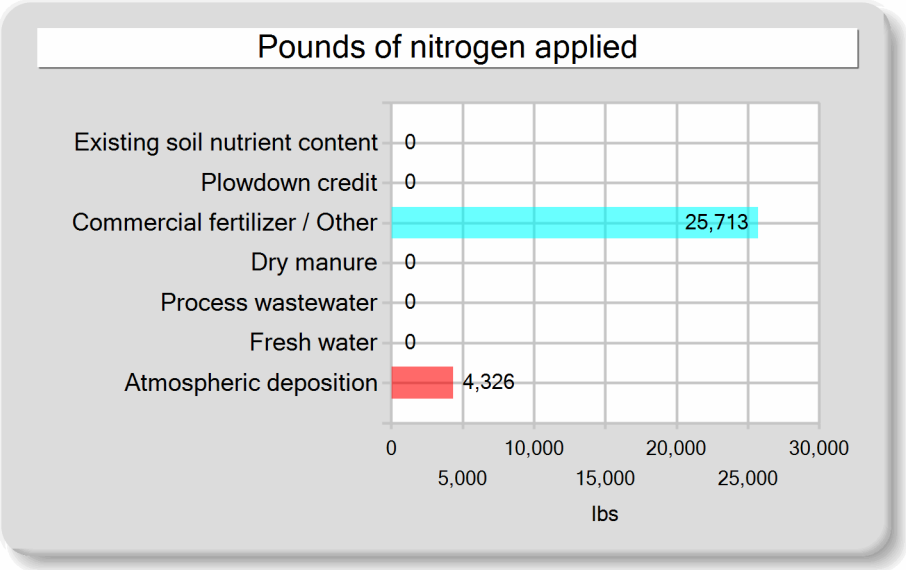
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

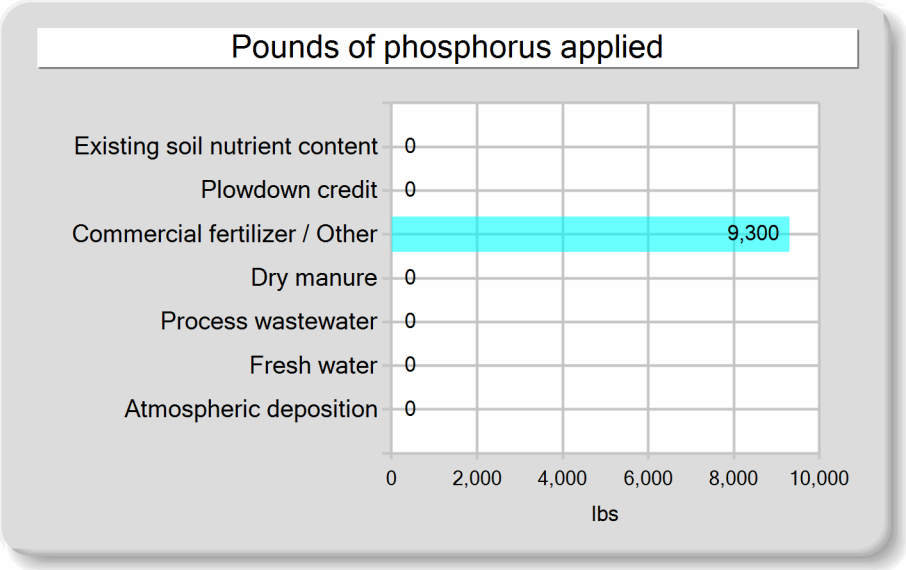
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25,712.54	9,300.00	13,485.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	309,208.45
Atmospheric deposition	4,326.00	0.00	0.00	0.00
Total nutrients applied	30,038.54	9,300.00	13,485.00	309,208.45
Anticipated crop nutrient removal	108,290.50	12,727.50	83,522.10	0.00
Actual crop nutrient removal	97,485.09	9,932.79	75,031.57	757,668.04
Nutrient balance	-67,446.55	-632.79	-61,546.57	-448,459.59
Applied to removed ratio	0.31	0.94	0.18	0.41

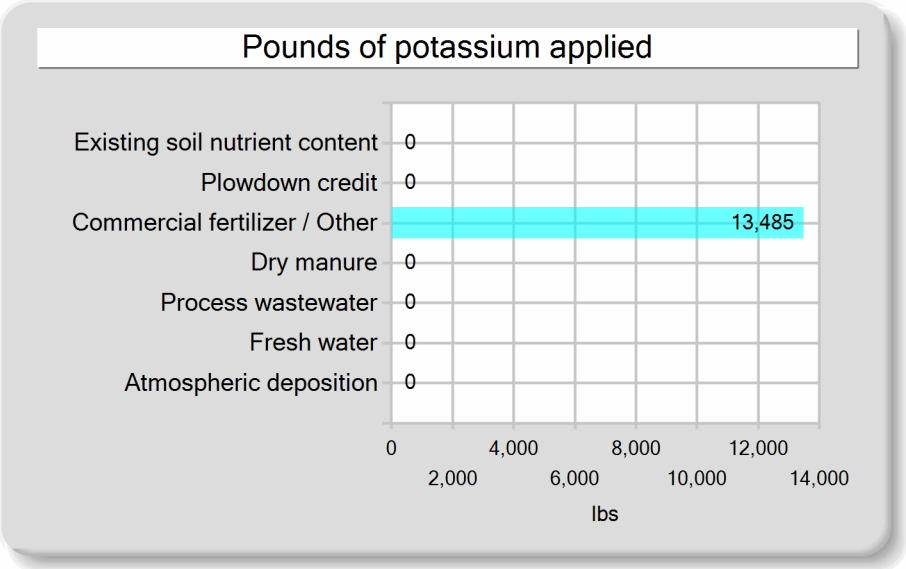
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

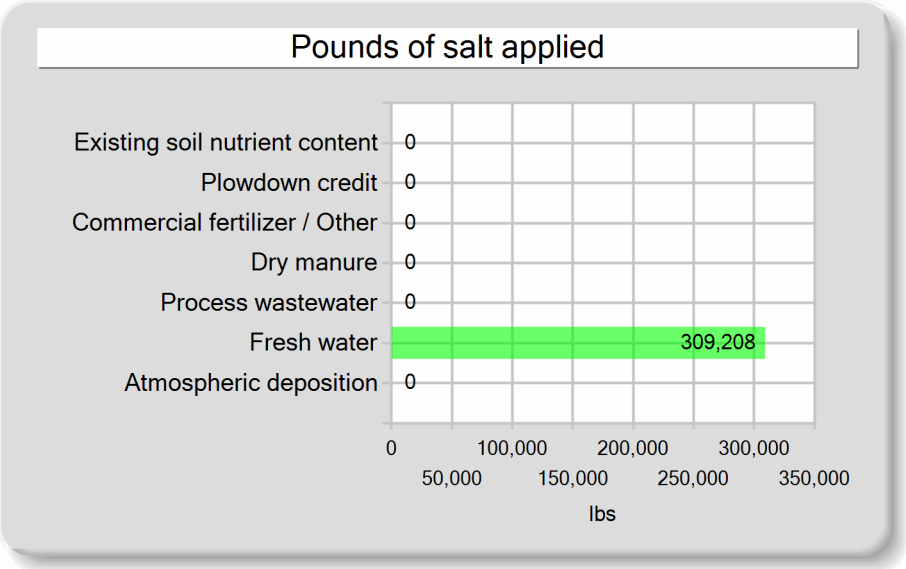


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No solid manure or wastewater was applied on the dairy fields for the 2023 crop year. Corrals had a larger cleaning and all solid manure was exported. Wastewater applications did not occur due to the small number of animals onsite and the discontinuation of milkbarn water ; wastewater instead evaporated in the storage.

Wastewater Transports from the feedlot occurred in the beginning of 2023 due to heavy rainfall:
1/5/23: 60,000 gallons
1/6/23: 72,000 gallons
1/7/23: 80,000 gallons
1/10/23: 60,000 gallons
1/11/23: 72,000 gallons
1/16/23: 60,000 gallons
1/17/23: 80,000 gallons
1/20/23: 40,000 gallons
1/24/23: 60,000 gallons
Additional freshwater diversions were completed in 2023 and have improved the amount of runoff going into the storages.

Diesel Well was not operational in 2023 and could not be sampled. It is now operational its sample results will be included in the 2024 Annual Report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Greg and Lisa O'Banion (Jun 26, 2024 16:49 PDT)	Greg and Lisa O'Banion (Jun 26, 2024 16:49 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Greg and Lisa O'Banion	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/26/2024	06/26/2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: O'Banion Ranches

Name of Dairy Facility: O'Banion Ranches

Facility Address:

15775 S Indiana RD	Dos Palos	Merced	93620
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Greg and Lisa O'Banion (209) 587-1419
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Azevedo Brothers Trucking

Address of Hauling Company/Person:

6445 W Hwy 152	Dos Palos	CA	93620
Number and Street	City	State	Zip Code

Contact Person: Franklin Azevedo (209) 777-4298
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Curran Ranch (209) 587-1446
Name Phone Number

15735 Indiana RD	Dos Palos	CA	93620
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

15735 Indiana RD	Dos Palos	93620
Address	City	Zip Code
		Merced
Street and nearest cross street (if no address)		County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,475.00 tons

Manure Solids Content: 76.8 %

Method used to determine amount of manure:

Total Loads Multiplied by Average Load Weight.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator Signature — Date 6/19/24

Hauler Signature — Date 6/21/24

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: O'Banion Ranches

Name of Dairy Facility: O'Banion Ranches

Facility Address:

15775 S Indiana RD	Dos Palos	Merced	93620
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Greg and Lisa O'Banion (209) 587-1419
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Azevedo Brothers Trucking

Address of Hauling Company/Person:

6445 W Hwy 152	Dos Palos	CA	93620
Number and Street	City	State	Zip Code

Contact Person: Franklin Azevedo (209) 777-4298
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Curran Ranch (209) 587-1446
Name Phone Number

15735 Indiana RD	Dos Palos	CA	93620
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

15735 Indiana RD	Dos Palos	93620
Address	City	Zip Code
		Merced
Street and nearest cross street (if no address)		County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 525.00 tons

Manure Solids Content: 57.4 %

Method used to determine amount of manure:

Total Loads Multiplied by Average Load Weight.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator Signature Date 6/19/24

Hauler Signature Date 6/21/24

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
O'Banion Ranches
15735 Indiana Rd.
Dos Palos, CA 93620

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 12:45 PM
SUBMITTED BY:
ANALYZED DATE: 6/7/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well-Rocha	T3158055A	Electrical Conductivity (EC)	EPA 120.1 and	0.977	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	178	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	102	mg/L
		Sodium (Na)	EPA 200.7	148	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	13.3	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	470	mg/L
		Calcium (Ca)	EPA 200.7	35.1	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	143	mg/L
		Soluble Salts (SALT-SOL)	Calculation	625	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)			2.37
Calcium (Ca)	94.7					
Chloride (Cl)	104	113 (T3158049A)	103 (T3158049A)			9.10
Magnesium (Mg)	98.5					
Nitrate Nitrogen (NO3-N)	105	112 (T3158049A)	109 (T3158049A)			2.08
Sodium (Na)	101					
Sulfate (SO4)	105	57.1 (T3158049A)	111 (T3158049A)			64.2
Total Dissolved Solids (TDS)	97.5			525 (T3158055E)	550 (Dup)	4.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
O'Banion Ranches
15735 Indiana Rd.
Dos Palos, CA 93620

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 12:45 PM
SUBMITTED BY:
ANALYZED DATE: 6/7/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Feedlot 1	T3158055B	Electrical Conductivity (EC)	EPA 120.1 and	2.08	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	260	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	185	mg/L
		Sodium (Na)	EPA 200.7	186	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	47.9	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.323	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1190	mg/L
		Calcium (Ca)	EPA 200.7	159	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	514	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1330	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)			2.37
Calcium (Ca)	94.7					
Chloride (Cl)	104	113 (T3158049A)	103 (T3158049A)			9.10
Magnesium (Mg)	98.5					
Nitrate Nitrogen (NO3-N)	105	112 (T3158049A)	109 (T3158049A)			2.08
Sodium (Na)	101					
Sulfate (SO4)	105	57.1 (T3158049A)	111 (T3158049A)			64.2
Total Dissolved Solids (TDS)	97.5			525 (T3158055E)	550 (Dup)	4.65

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Compliance Analysis Report

Customer
O'Banion Ranches
15735 Indiana Rd.
Dos Palos, CA 93620

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 12:45 PM
SUBMITTED BY:
ANALYZED DATE: 6/7/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Feedlot 2	T3158055C	Electrical Conductivity (EC)	EPA 120.1 and	0.266	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	94.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	< 0.600	mg/L
		Sodium (Na)	EPA 200.7	45.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	1.07	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	120	mg/L
		Calcium (Ca)	EPA 200.7	10.4	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	20.5	mg/L
		Soluble Salts (SALT-SOL)	Calculation	170	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)			2.37
Calcium (Ca)	94.7					
Chloride (Cl)	104	113 (T3158049A)	103 (T3158049A)			9.10
Magnesium (Mg)	98.5					
Nitrate Nitrogen (NO3-N)	105	112 (T3158049A)	109 (T3158049A)			2.08
Sodium (Na)	101					
Sulfate (SO4)	105	57.1 (T3158049A)	111 (T3158049A)			64.2
Total Dissolved Solids (TDS)	97.5			525 (T3158055E)	550 (Dup)	4.65

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Compliance Analysis Report

Customer
O'Banion Ranches
15735 Indiana Rd.
Dos Palos, CA 93620

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 12:45 PM
SUBMITTED BY:
ANALYZED DATE: 6/7/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Shop Well	T3158055D	Electrical Conductivity (EC)	EPA 120.1 and	0.804	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	182	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	69.2	mg/L
		Sodium (Na)	EPA 200.7	127	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	10.6	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	395	mg/L
		Calcium (Ca)	EPA 200.7	27.0	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	114	mg/L
		Soluble Salts (SALT-SOL)	Calculation	515	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)			2.37
Calcium (Ca)	94.7					
Chloride (Cl)	104	113 (T3158049A)	103 (T3158049A)			9.10
Magnesium (Mg)	98.5					
Nitrate Nitrogen (NO3-N)	105	112 (T3158049A)	109 (T3158049A)			2.08
Sodium (Na)	101					
Sulfate (SO4)	105	57.1 (T3158049A)	111 (T3158049A)			64.2
Total Dissolved Solids (TDS)	97.5			525 (T3158055E)	550 (Dup)	4.65

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Compliance Analysis Report

Customer
O'Banion Ranches
15735 Indiana Rd.
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Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 12:45 PM
SUBMITTED BY:
ANALYZED DATE: 6/7/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy	T3158055E	Electrical Conductivity (EC)	EPA 120.1 and	1.05	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	180	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	112	mg/L
		Sodium (Na)	EPA 200.7	118	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	24.4	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	525	mg/L
		Calcium (Ca)	EPA 200.7	53.0	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	161	mg/L
		Soluble Salts (SALT-SOL)	Calculation	674	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)			2.37
Calcium (Ca)	94.7					
Chloride (Cl)	104	113 (T3158049A)	103 (T3158049A)			9.10
Magnesium (Mg)	98.5					
Nitrate Nitrogen (NO3-N)	105	112 (T3158049A)	109 (T3158049A)			2.08
Sodium (Na)	101					
Sulfate (SO4)	105	57.1 (T3158049A)	111 (T3158049A)			64.2
Total Dissolved Solids (TDS)	97.5			525 (T3158055E)	550 (Dup)	4.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: O'Banion Ranches
Mailing address: 15735 Indiana Rd
(not facility address) Dos Palos, CA 93620
Phone#: (209) 587-1446

Authorized Copy Released To: Kristy Rocha

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	6/7/23	10:00 am	Well-Rocha	Water	20.2°		X				H14
2	6/7/23	10:00am	Feedlot 1	Water	20.6		X				
3	6/7/23	10:05am	Feedlot 2	water	20.0°		X				
4	6/7/23	10:03am	Shop Well	water	20.0°		X				
5	6/7/23	10:10am	Dairy	water	20.0°		X				
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]					
2nd	[signature]	DENELE ANALYTICAL, INC.	6-7-23 12:45			T3158055

Required to be submitted to any agency? (YES / NO)